Environment Setup Guide

WSD4302 — Web Services Development (Java EE / Jakarta EE)

Lab 1 preparation — install and verify all required tools on your own laptop before class

This guide walks you through installing every tool you will need for this course, starting from a completely clean Windows computer. Follow the steps in order — later tools (Maven) depend on earlier ones (JDK) being installed correctly. Do this before your first practical lesson; class time will be used for HTTP practice, not installation.

What you will install

- Eclipse Temurin JDK 21 (LTS)
- Apache Maven 3.9 or newer
- Git
- Bruno (API client)
- curl (built into Windows 10/11 — just needs verifying)
- IntelliJ IDEA Ultimate (free with a student license)

1. Install JDK 21 (Eclipse Temurin)

Step 1. Download the installer

Go to adoptium.net and download the Windows installer (.msi) for Temurin 21 (LTS), x64. Choose the .msi installer, not the .zip archive — it configures everything automatically.

Official link: adoptium.net/temurin/releases/?version=21

On that page, filter by Operating System: Windows, Architecture: x64, Package Type: JDK, then click the .msi download button.

Step 2. Run the installer

Run the downloaded .msi file. Accept the defaults, but on the "Custom Setup" screen make sure these options are enabled (they are usually off by default):

- Set JAVA_HOME variable
- JavaSoft (Oracle) registry keys
- Add to PATH

Enabling these now saves you from manually editing environment variables later.

Step 3. Verify

Close any open terminal windows. Open a new Command Prompt (cmd) and run:

```
java -version
javac -version
```

Both commands must report version 21. If "javac" is not found, you installed a JRE instead of a JDK — uninstall and download the JDK package instead.

2. Install Apache Maven 3.9+

Unlike the JDK, Maven does not come with an installer — it is a single zip archive that you place manually and point Windows to.

Step 1. Download

Go to maven.apache.org → Download, and download the "Binary zip archive" (e.g. apache-maven-3.9.16-bin.zip). Do not download the "Source zip" — that is source code, not a runnable program.

Official link: maven.apache.org/download.cgi

On that page, under "Apache Maven 3.9.16", use the row labeled "Binary zip archive" — not tar.gz (Linux/macOS), not any "Source" row, and not the Preview 3.10/4.0 sections.

Step 2. Extract to a permanent folder

Extract the zip to a stable location outside Downloads — Windows sometimes double-nests the folder when you use "Extract All", and Downloads gets cleaned out periodically. A simple path avoids both problems, for example:

```
C:\maven\apache-maven-3.9.16
```

Check that this folder directly contains a bin subfolder with mvn.cmd inside it.

Step 3. Set environment variables

Press Win + R, type sysdm.cpl, press Enter → Advanced tab → Environment Variables.

In the TOP section — "User variables for <you>" — not the bottom "System variables" section (System variables need administrator rights and can silently fail to save without one):

- Click New… under User variables. Name: MAVEN_HOME. Value: the folder from Step 2 (e.g. C:\maven\apache-maven-3.9.16).
- Select the Path variable under User variables → Edit… → New → add: %MAVEN_HOME%\bin

Click OK on every dialog box to save — clicking Cancel anywhere discards the change.

Step 4. Verify

Close ALL open terminal and IDE windows (they cache the old PATH and will not see the update). Open a brand-new Command Prompt and run:

```
mvn -v
```

You should see the Maven version and, importantly, the Java version it is using — confirm it says version 21, matching your JDK.

> **⚠ Common error**
> "'mvn' is not recognized as an internal or external command" almost always means one of two things:
> 1) You edited System variables instead of User variables and it didn't have permission to save, or
> 2) You still have an old terminal window open from before the change. Close everything and open a fresh terminal.

3. Install Git

Step 1. Download and install

Go to git-scm.com/download/win and download the 64-bit installer. Run it and accept the defaults throughout — the standard installation adds Git to PATH automatically and includes Git Bash.

Official link: git-scm.com/download/win

This page starts the correct 64-bit download automatically as soon as you open it.

Step 2. Configure your identity

Open a new terminal and set your name and email — Git attaches these to every commit you make:

```
git config --global user.name "Your Name"
git config --global user.email "you@example.com"
```

Step 3. Verify

```
git --version
```

4. Install Bruno

Bruno is a free, offline API client (an alternative to Postman) — no account or sign-in is required.

Step 1. Download and install

Go to usebruno.com → Download, get the Windows installer, and run it with default settings.

Official link: usebruno.com/downloads

The site auto-detects your OS and recommends "Windows x64 (.exe)" — that is the one to click.

Step 2. Open it once

Launch Bruno and confirm it opens to the "My Workspace" screen. Nothing else to configure yet — we will create a collection together in class.

5. Verify curl

curl is already built into Windows 10 and 11, so nothing to install — just confirm it and learn one important trap.

Step 1. Verify

```
curl --version
```

This should print a version number (8.x) and a list of supported protocols. If it fails, restart your computer — a Windows update sometimes needs a reboot to register the built-in curl.

> **⚠ PowerShell trap**
> In PowerShell, the plain word "curl" is secretly an alias for Invoke-WebRequest, a completely different tool with different flags — your -i, -X, -d options will not behave the same way.
> Always type curl.exe explicitly in PowerShell, or simply use the regular Command Prompt (cmd) instead of PowerShell for this course. Command Prompt does not have this alias problem.

6. Install IntelliJ IDEA Ultimate

Step 1. Get a student license

Go to jetbrains.com/community/education/#students and apply for a free student license using your university email address.

Official link: jetbrains.com/community/education/#students

Step 2. Download and install

Once approved, download IntelliJ IDEA Ultimate (not the free Community edition — Ultimate is required for the Jakarta EE / web-application tooling used in this course) from jetbrains.com/idea and run the installer with default settings.

Official link: jetbrains.com/idea/download (Windows)

Make sure "Ultimate" is selected on that page, not "Community".

Step 3. Sign in

On first launch, sign in with your JetBrains account to activate the Ultimate license.

7. Final checklist

Open one fresh Command Prompt window and run all of these in sequence. Every one should succeed with no "not recognized" errors:

```
java -version
javac -version
mvn -v
git --version
curl --version
```

If every command above prints a version number, your machine is fully ready for Lab 1. Bring this laptop, already set up, to your next practical lesson.

Getting help

If you get stuck on any step, take a screenshot of the exact error message and bring it to class (or send it ahead of time) — most setup issues are one of the three problems covered in the warning boxes above: wrong variable section, a stale terminal window, or the PowerShell curl alias.